Republic of the Philippines
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
Ifugao Second District Engineering Office
Talite, Aguinaldo, Ifugao, Philippines

BAGONG PILIPINAS

PRELIMINARY GEOHAZARD ASSESSMENT REPORT FOR THE FY 2026 HIGHWAY PROJECT

ASSET PRESERVATION PROGRAM-CONSTRUCTION/ UPGRADING/ REHABILITATION OF DRAINAGE ALONG NATIONAL ROADS - SECONDARY ROADS - BANAUE-MAYOYAO-A LISTA-ISABELA BDRY RD - K0428 +793 - K0429 + 537, K0431 + 000 - K0431 + 052, K0431 + 161 - K0431 + 233, K0431 + 300 - K0431 + 322, K0431 + 409

June 2025

Design Guidelines, Criteria and Standards: Volume 2A – GeoHazard Assessment

Preliminary Geohazard Assessment

Project Summary Description	The site covered by this geohazard assessment is a Highway project in Alfonso Lista, Ifugao.				
Location	**Northing/Lat:**	**16°54'02.80"N**	**Easting/Long:**	**121°27'25.66"E**	**Lat/Long WGS 1984 (Google Earth Lat/Long)**

For each item below except Seismic Hazard, mark ONE column with a tick or ring an option.

Hazard	**Section Reference**	**(1) No hazard exists. No further action required.**	**(2) Hazard exists but can be addressed and mitigation measures are included in the project budget.**	**(3) Hazard exists. Future survey and site investigation must include for further identification and mitigation of the hazard.**	**(4) Specialist Expertise Required, or consider relocating the project site.**
Seismic	2.2	**Distance to Active Fault**	15820 meters	**Peak Ground Acceleration (Uncorrected)**	0.450 g
Liquefiable Soils	2.3	2			
Volcanic	2.4	2			
Unstable Slopes	2.5	3			
Karst	2.6	1			
Mining Activities	2.7	1			
Expansive soils	2.8.1	1			
Fills	2.8.3	(Refer to "Contaminated Soils")			
High Compressibility Soils	2.8.5	1			
Contaminated Soils	2.8.7	1			
Groundwater	2.9	(Refer to "Unstable Slopes")			
Flooding, scour & erosion	2.1	1			
Tsunamis, seiches, storm surges	2.11	1			
Geothermal activity	2.12	1			
Overall Classification	3				

Mark the rightmost column for which there is a tick above

I. PROJECT DESCRIPTION / BACKGROUND

Project Name:	ASSET PRESERVATION PROGRAM-CONSTRUCTION/ UPGRADING/ REHABILITATION OF DRAINAGE ALONG NATIONAL ROADS - SECONDARY ROADS - BANAUE-MAYOYAO-A LISTA-ISABELA BDRY RD - K0428 +793 - K0429 + 537, K0431 + 000 - K0431 + 052, K0431 + 161 - K0431 + 233, K0431 + 300 - K0431 + 322, K0431 + 409
Location:	Alfonso Lista, Ifugao
District Engineering Office (DEO):	Ifugao Second District Engineering Office
Geographic Coordinates:	
- WGS84	16°54'02.80"N 121°27'25.66"E
- PRS92 Philippine Zone 3	548,562.44E, 1,869,357.03N
Scope of Work:	Construction of Drainage and Slope Protection Structures
Project Cost:	Php. 20,982,000.00

Reference:

DPWH Design Guidelines, Criteria and Standards (DPWH DGCS, 2015)

DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)

Prepared by:

SIGNED

EDWARD A. CARIÑO

Chief, Planning and Design Section

II. DISCUSSION ON GEOLOGY

Regional Geology:	Central Cordillera – Eastern Cordillera Geological Province
Rock Type:	Upper Miocene-Pliocene (Sedimentary & Metamorphic Rocks
Soil Type:	Silty Clayey (A-2-4)
Thickness:	3.0 to 10.0 meters
Geologic Structures:	N/A

Discussions:

Please refer to the Geology of the Philippines

Geologic Structures

*Geologic structure is one of the most important factors in slope failure. These include **joints and faults**, which are critical in slopes. These are the zones of weaknesses of rocks and affect the hydrologic condition of slopes. The nature, strength and orientation of fractures (joints/faults) are the main factors that affect the slope condition and determine the type of slope failure (DPWH).*

References:

DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)

III. DISCUSSION ON SEISMICITY

Active Faults		
Nearest Active Faults	Length of Fault (km) (see below)	Distance from Site (km) (Measured from QGIS or Google Earth)
1. East Cordillera Fault (Santiago Segment)	51.00	8.10
2. Philippine Fault (Hapap Segment)	71.00	44.85
3. Dacalan Fault	19.00	45.90

Discussions:

Active faults are faults that have moved within the last 10,000 years. Ground rupture is a displacement along an active fault trace that reaches the surface. The ***recommended buffer zone****, or Zone of Avoidance, against ground rupture hazard is* ***at least 5 meters*** *on both sides of the active fault or from its zone of deformation (PHIVOLCS).*

Reference:
Wells and Coppersmith (1994) equation
Fukushima and Tanaka (1990) equation

Historical Records of Seismicity (M > 6.0)		
Date	Maximum Magnitude	Distance (km)
1. 07/22/2022	7.00	110.79
2. 10/25/2022	6.70	112.26
3. 10/27/2022	6.40	121.74

Discussions:

An active fault may show evidence or may have documented history of recent movements.

For Roads and Flood Control Projects only Direct Seismic Hazard Analysis (DSHA)				
Soil Profile Type from NHERP Scheme	Deterministic Peak Ground Acceleration (PGA) in g	PGA (Corrected) in g	Seismic Horizontal Coefficient, Kh	Seismic Vertical Coefficient, Kv
SD	0.45	0.39	0.20	0.10

Discussions:

All sites may be affected by ground shaking in the event of an earthquake and ***can be mitigated*** *by following the provisions of the National Building Code and the National Structural Code of the Philippines NSCP), 2015.*

References:

Department of Science and Technology–Philippine Institute of Volcanology and Seismology (DOST–PHIVOLCS), Peak Ground Acceleration Map of the Philippines, 500-Year Return Period on Stiff Soil, Philippine Earthquake Model (PEM), 2017.

Hynes-Griffin and Franklin (1984) equation

DPWH Design Guidelines, Criteria and Standards (DPWH DGCS, 2015)

V. DISCUSSION ON GEOMORPHOLOGY, TOPOGRAPHY AND CLIMATE

Geomorphology:	Hills
Topography:	Hillslope
Climate:	Type III

Discussions:

The site is underlain by a slope terrain wherein the expected geomorphological processes include rotational slides, and debris slides. Although the detailed geohazard assessment will be supplemented and verified with the study of existing geohazard maps and site reconaissance.

Topography and Geomorphology

Landforms: Hills (Hill Slope). The project site is characterized by hilly terrain with sloping ground surfaces, which may increase susceptibility to rainfall-induced slope instability, landslides, and soil erosion, particularly during prolonged or intense rainfall. Water infiltration can increase pore-water pressure and reduce soil shear strength, while concentrated surface runoff may cause rill and gully erosion. Slope cutting, excavation, and inadequate drainage may further aggravate these hazards. Therefore, proper surface drainage, erosion control, slope protection, and regular monitoring are recommended to minimize potential geohazard impacts.

Climate

Type III: No very pronounced maximum rain period, with a dry season lasting only from one to three months, either during the period from December to February or from March to May. This type resembles Type I since it has a short dry season (PAG-ASA).

References:

Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAG-ASA)
DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)

VI. DISCUSSION ON GEOHAZARD SUSCEPTIBILITY

Volcanic Hazard:	Nearest Active Volcano Approximately 162.1 km southwest of Cagua Lahar: Safe Ash fall: Prone

Discussions:

Active volcanoes are those that erupted within historical times (within the last 600 years). Accounts of these eruptions were documented by man within the last 10,000 years based on the analyses of material from young volcanic deposits.

Lahar

Lahars (an Indonesian term), sometimes called volcanic mudflows or debris flows, are slurries of volcanic sediment, debris and water that cascade down a volcano's slopes through rivers and channels. Lahars in tropical areas are mainly generated by torrential rainfall on unconsolidated deposits from a past eruption (PHIVOLS).

Ashfall

In case of future eruptions, the site may be affected by ash fallout, depending on the scale of eruption and prevailing wind direction at the time of eruption. Generally, ashfall is heavier near the active vent and thins out indefinitely away from the eruption center.

References:

Philippine Institute of Volcanology and Seismology (PHIVOLCS)

HazardHunterPH

Rainfall-Induced Landslide Map:	***Moderate Susceptibility: Possible landslide occurrence***

Discussions:

Moderate Susceptibility: Possible landslide occurrence

Areas with moderate susceptibility to rain-induced landslides are those with moderately steep slopes where soil creep and other indications of possible landslide occurrence are present.

Implementation of appropriate mitigation measures as deemed necessary by project engineers and LGU building officials is recommended for landslide-susceptible areas. This includes performing site-specific studies to address potential foundation/slope stability problems.
Monitoring of signs/evidences of ground movement such as tension cracks, tilted trees and fences, and bulging road sections in areas that are moderately to critically susceptible to landslides should be done regularly and reported to local authorities and/or the MGB.

References:
Mines and Geosciences Bureau (MGB)

Earthquake-Induced Landslide	Prone; Moderate Susceptibility

Discussions:

Earthquake-induced landslides are the downward slope movement of rocks, solid and other debris commonly triggered by strong shaking. Avoidance is recommended for sites with earthquake-induced landslide hazard unless appropriate engineering interventions are in place.

References:
Philippine Institute of Volcanology and Seismology (PHIVOLCS)

HazardHunterPH

Flood Hazards and Fluvial Hazards	*Low Susceptibility: 0.5 to 1.0 meter flood height and/or 1 to 3 days flooding*

Discussions:
Low Susceptibility: less than 0.5 meters flood height and/or less than 1 day flooding
Areas with low susceptibility to floods are likely to experience flood heights of less than 0.5 meters and/or flood duration of less than 1 day. These include low hills and gentle slopes that have sparse to moderate drainage density.

The implementation of appropriate mitigation measures as deemed necessary by project engineers and LGU building officials is recommended for areas that are susceptible to various flood depths. Site-specific studies including the assessment for other types of hazards should also be conducted to address potential foundation problems (MGB).

Based on the field, there are indications of Channel Bank erosion due to increased channel discharge, channel confinement, and channel migration. Further, there are gullies and presence of gravels, cobbles, boulders in the stream bed, which are indications of either Debris Flows or Channel Bed Scour (DPWH-GGA, 2017).
For Roads

An embankment with a height of at least 1 meter should be considered before the construction of concrete pavement. It should be noted that asphalt pavement is not recommended within areas with high susceptibility to flooding (DPWH-GGA, 2017).
References:
Mines and Geosciences Bureau (MGB)
DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)

Liquefaction Hazard	Safe

Discussions:
Liquefaction is a phenomenon wherein the ground, especially near the river, lake and coasts, behaves like liquid similar to quicksand due to very strong shaking.

References:
Idriss and Boulanger (2014)
Philippine Institute of Volcanology and Seismology (PHIVOLCS)
National Structural Code of the Philippines (NSCP, 2015)
DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)
DPWH Bridge Seismic Design Specifications (BSDS, 2013)
Terzaghi and Peck (1967)
HazardHunterPH

Tsunami, Seiches and Storm Surge Hazard	Safe

Discussions:
A tsunami is a series of sea waves commonly generated by under-the-sea earthquakes. Its threat to people's lives can be addressed by community preparedness and a tsunami evacuation plan. Advice for tsunami evacuation comes from public agencies and the local government. More importantly, coastal communities must learn to evacuate themselves when they recognize the three natural signs of tsunami, which are 1) strong ground shaking, 2) unusual rise or fall of sea level, and 3) strong or unusual sound coming from the sea (PHIVOLCS).

Storm Surge
A storm surge ("daluyong ng bagyo") is the abnormal rise in sea level that occurs during tropical cyclones or "bagyo". It happens when a very strong tropical cyclone blows-off excessive amounts of seawater toward low-lying coastal communities. It is catastrophic and life-threatening because a storm surge can cause massive inland flooding, sometimes in unimaginable heights. It is even more dangerous when the storm surge coincides with a high tide. For storm surge-prone communities, the most important considerations are 1) the strength of the tropical cyclone, 2) the height of the surge, and 3) if the community is located in a low-lying area (PAGASA).

References:
Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAG-ASA)
Philippine Institute of Volcanology and Seismology (PHIVOLCS)

Mining Activities	Safe

Discussions:
If there are mining-related depressions or ground/structure cracks or nearby mine openings/portals present in site, these may eventually result in collapse, which may damage the structures. Hence, it is recommended to conduct further investigation by geophysical exploration such as Gravity Survey or Ground Penetrating Radar (GPR) and then identify feasible mitigating measures or consider relocating / realigning the site.

Building
It should never be constructed in old mine shafts and drift as it will have the possibility of collapsing. Pile foundations are recommended in areas that are filled with embankment materials. Ground improvement is necessary to obtain a competent bearing layer especially in rehabilitated mine areas (DPWH-GGA, 2017).

References:
DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)

Karst Hazards	Safe (See attached Field Photographs)

Discussions:
Karst/Sinkholes
The word "karst" is a comprehensive term applied to limestone areas that possess a topography peculiar to and dependent upon underground solution and the diversion of surface waters to underground routes (DPWH-GGA, 2017).

Caves
Caves are biodiverse and sensitive environmental features which are protected by Philippine Law and DENR under R.A. 9072, and relocation of the site should be considered.

Roads
Before finalizing the route of the road alignment, a detailed geological/geotechnical survey must be conducted such as borehole drilling and geophysical survey (using Ground Penetrating Radar, GPR and/or Seismic Multichannel Analysis Surface Wave, MASW) to avoid areas with sinkhole problems. If there will be no option, cement grouting, coarse backfill, reinforced concrete slabs, and/or diversion of drainage away from sinkholes are options to mitigate areas with sinkholes (DPWH-GGA, 2017).

Buildings and Bridges
Borehole drilling is necessary to determine the presence of any void/sinkhole within the location of the piers and abutments. If there will be sinkholes along the proposed location of the piers or abutments, the plan should be reconstructed or cement grouting is an option, if it is economical (DPWH-GGA, 2017).

References:
Republic Act (R.A.) 9072
DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)

VII. DISCUSSION ON PROBLEMATIC SOILS

Poorly Compacted Fills	Safe

Discussions:

Poorly Compacted Fills
During construction stage, there are three (3) main types of fills, which are likely to be encountered: 1) Engineered Fills consist of certain materials compacted to provide a stable foundation for further development. Such fills should not be a Geohazard, except for possible liquefaction. 2) Un-engineered Fills refer to materials dumped without selection or compaction but excluding waste dumps or landfills. 3) Waste Dumps or Landfills refer to materials likely to be compressible and unsuitable for founding of infrastructure but not otherwise be considered a Geohazard (DPWH DGCS, 2015).

References:
DPWH Design Guidelines, Criteria and Standards (DPWH DGCS, 2015)

Contaminated Soils	Safe

Discussions:

Contamination of natural soils by industrial or related activities will result to the toxicity of the ground. Contaminants usually result from dumping of industrial wastes. Methane gas is generated during the decay process of organic materials in landfill, as well as naturally occurring in organic deposits such as coal. Any land used for industrial purposes should be assessed for potential contamination, which may be hazardous to users of the land (DPWH DGCS, 2015).

References:
DPWH Design Guidelines, Criteria and Standards (DPWH DGCS, 2015)

Expansive Soils	Safe

Discussions:
If the site contains soils with ***Plasticity Index (PI) greater than or equal to 35,*** *which by are considered Expansive Soils, mitigation measures should be considered in the design (DPWH-GGA, 2017).*

Roads
The removal/replacement of expansive soil is an option if the thickness is minimal. On the other hand, ground improvement such as compaction or mixing with non-expansive soils to lower the expansion potential to an acceptable layer can be done (DPWH-GGA, 2017).

References:
DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)
American Association of State Highway and Transportation Officials (AASHTO) T 258-81

Highly Compressible and Collapsible Soils	Safe

Discussions:

High compressibility soils require to be identified at an early stage of construction and development in order to achieve suitable period to be provided for the additional costs in the engineering design phase (DPWH DGCS, 2015).

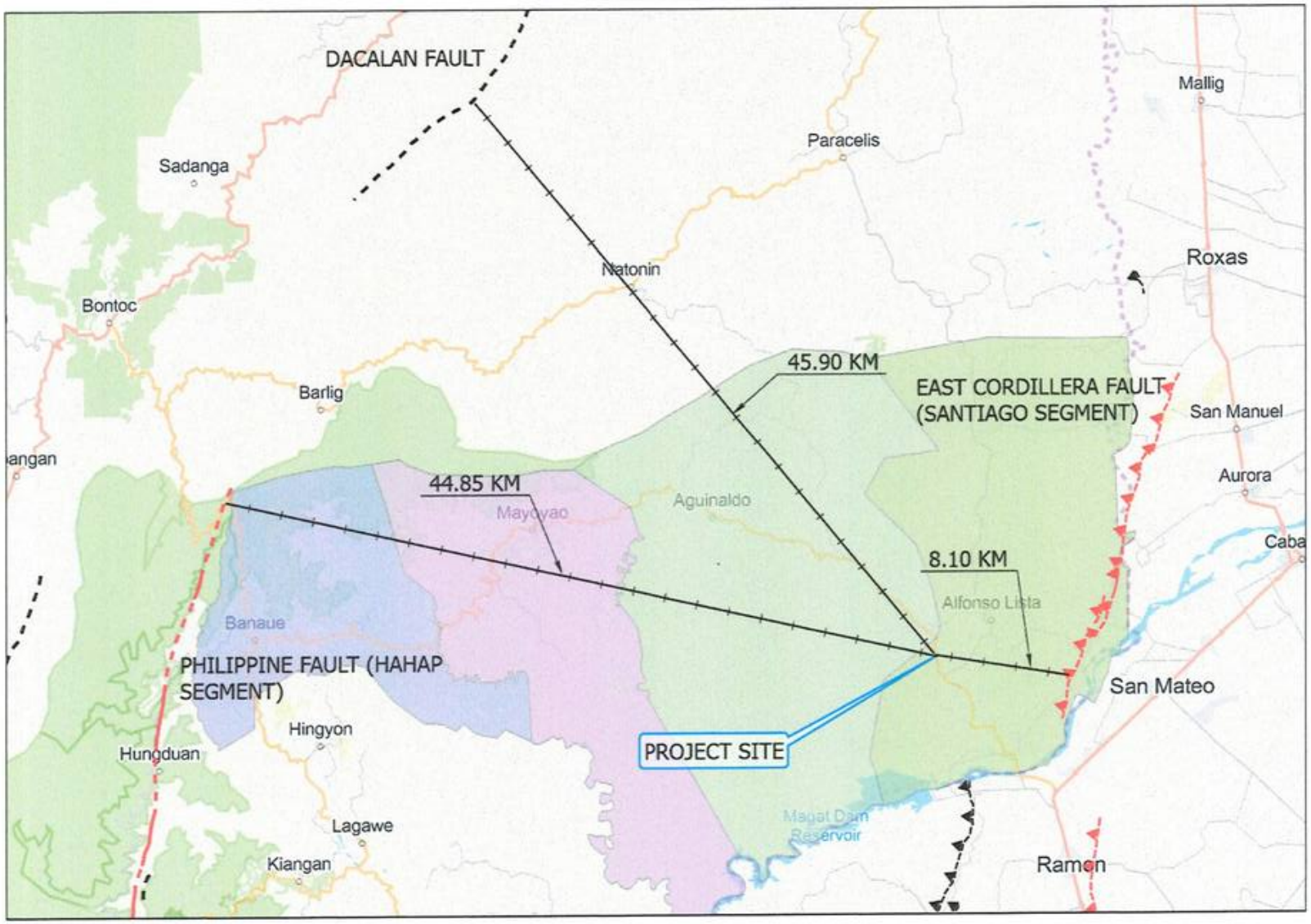
The two (2) most common types of highly compressible soils are peat and recent soft clay. These are both found mainly in low-lying areas, adjacent to sea level in river basins with meandering rivers. Peat is also found in higher and elevated terrain with depressions

If the site contains ***soils with SPT-N values less than or equal to 4****, which are considered Soft or Highly Compressible Soils, which may be susceptible to Settlement, mitigation measures should be considered in the design. These include removal/replacement, ground improvement, prefabricated vertical drains, and others as deemed applicable/economical (DPWH-GGA, 2017).*

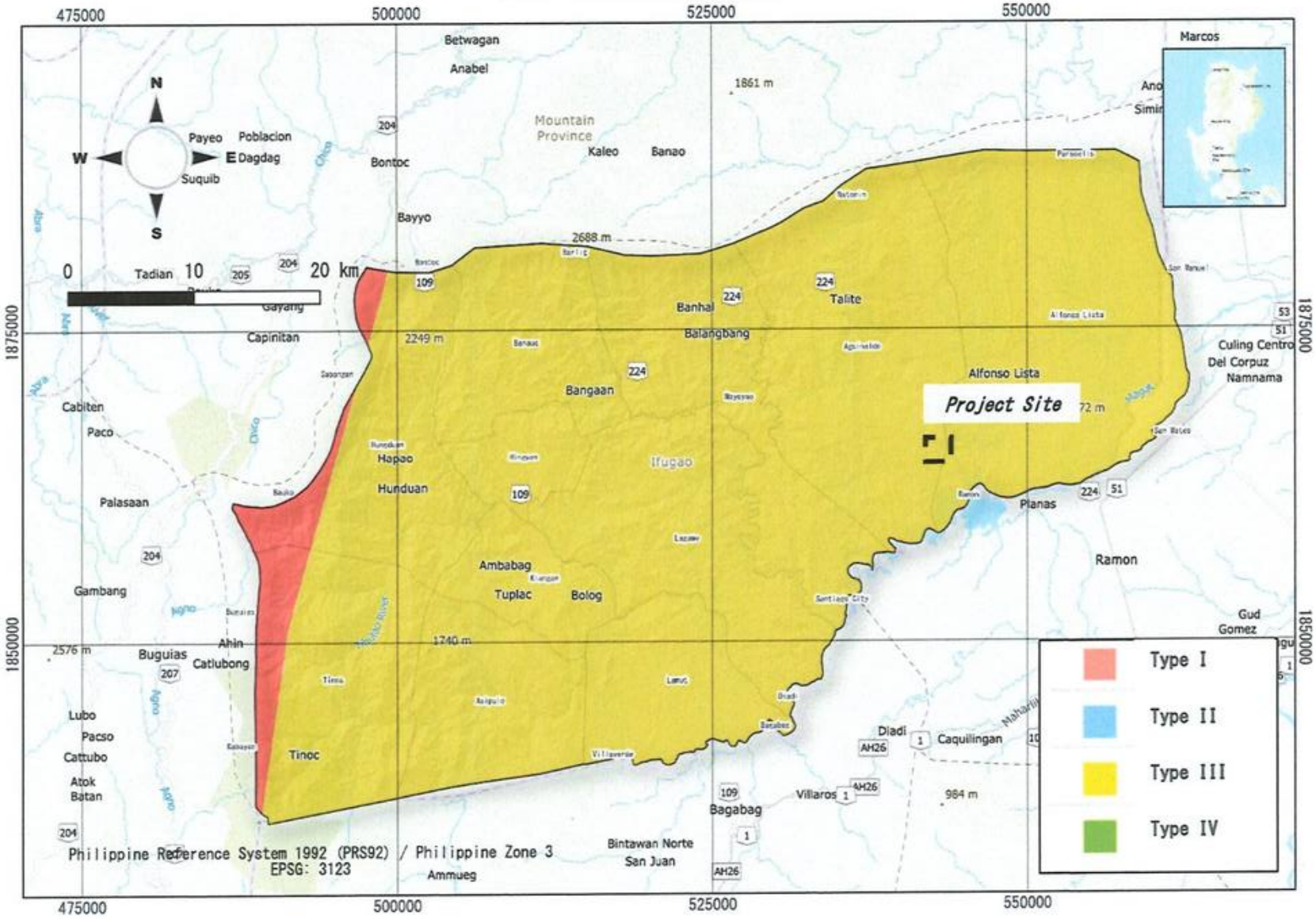
References:

DPWH Design Guidelines, Criteria and Standards (DPWH DGCS, 2015)
DPWH Guidelines on Geohazard Assessment Volume I-III (DPWH-GGA, 2017)

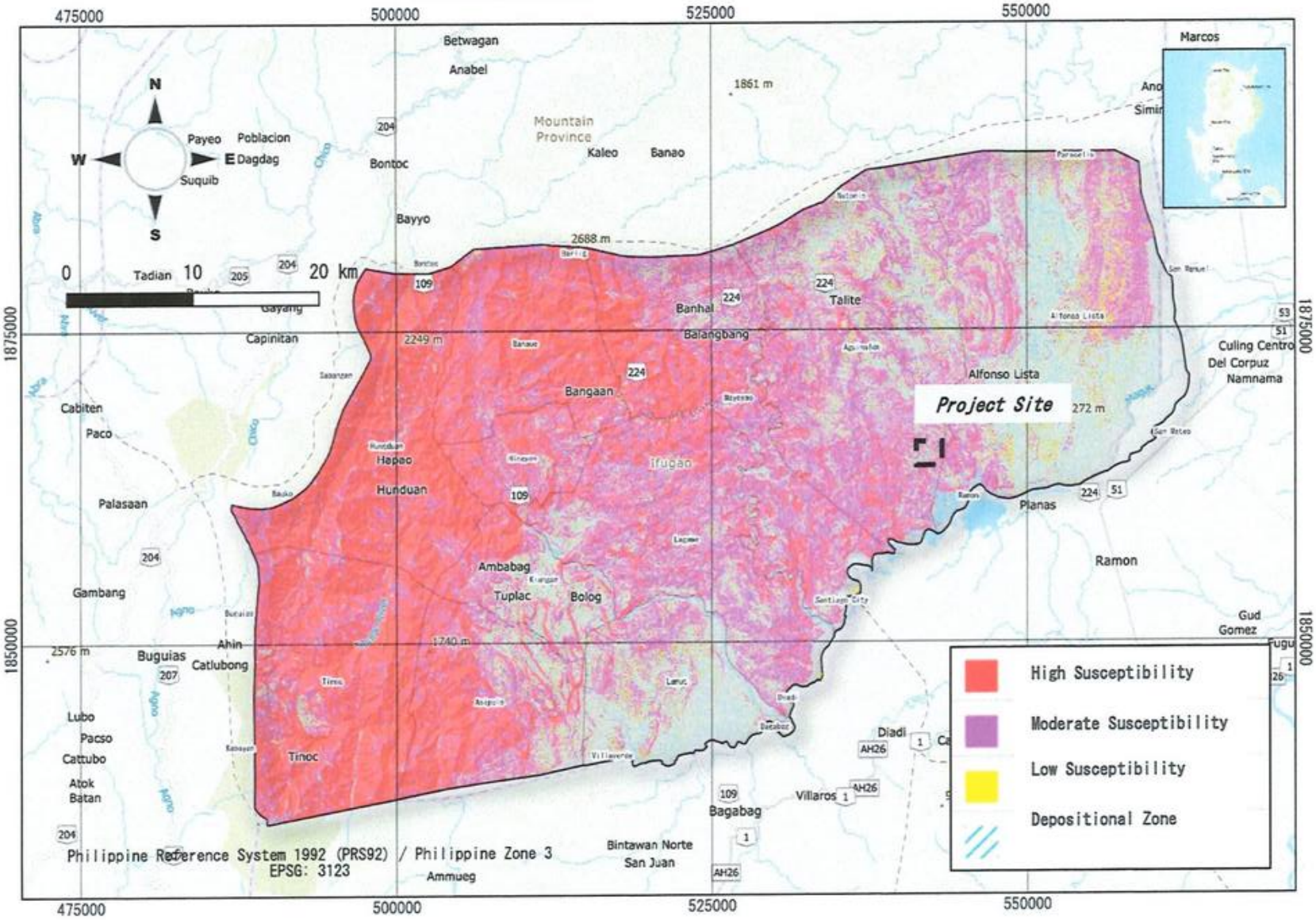
Active Fault Map



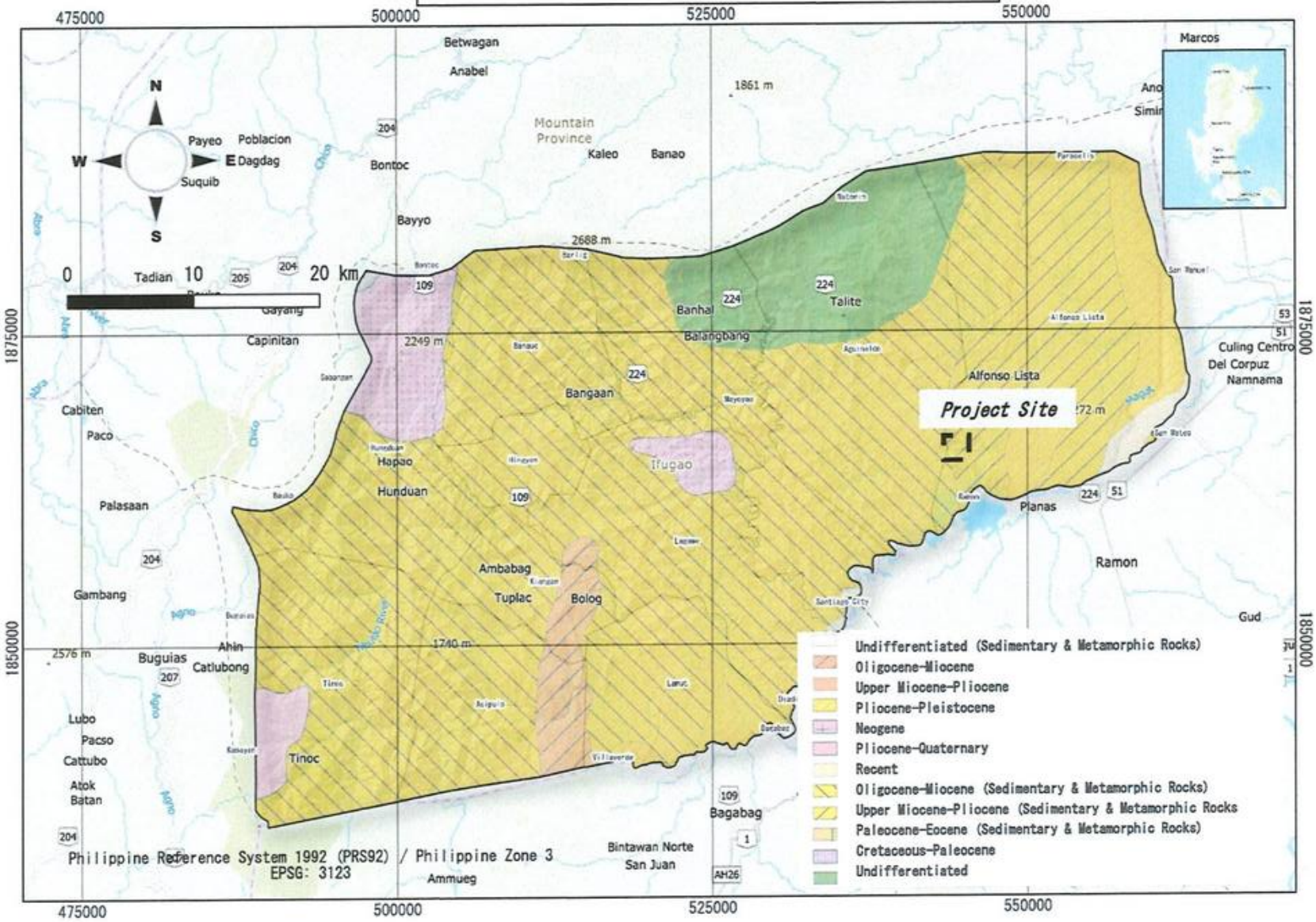
Climate Map of Ifugao



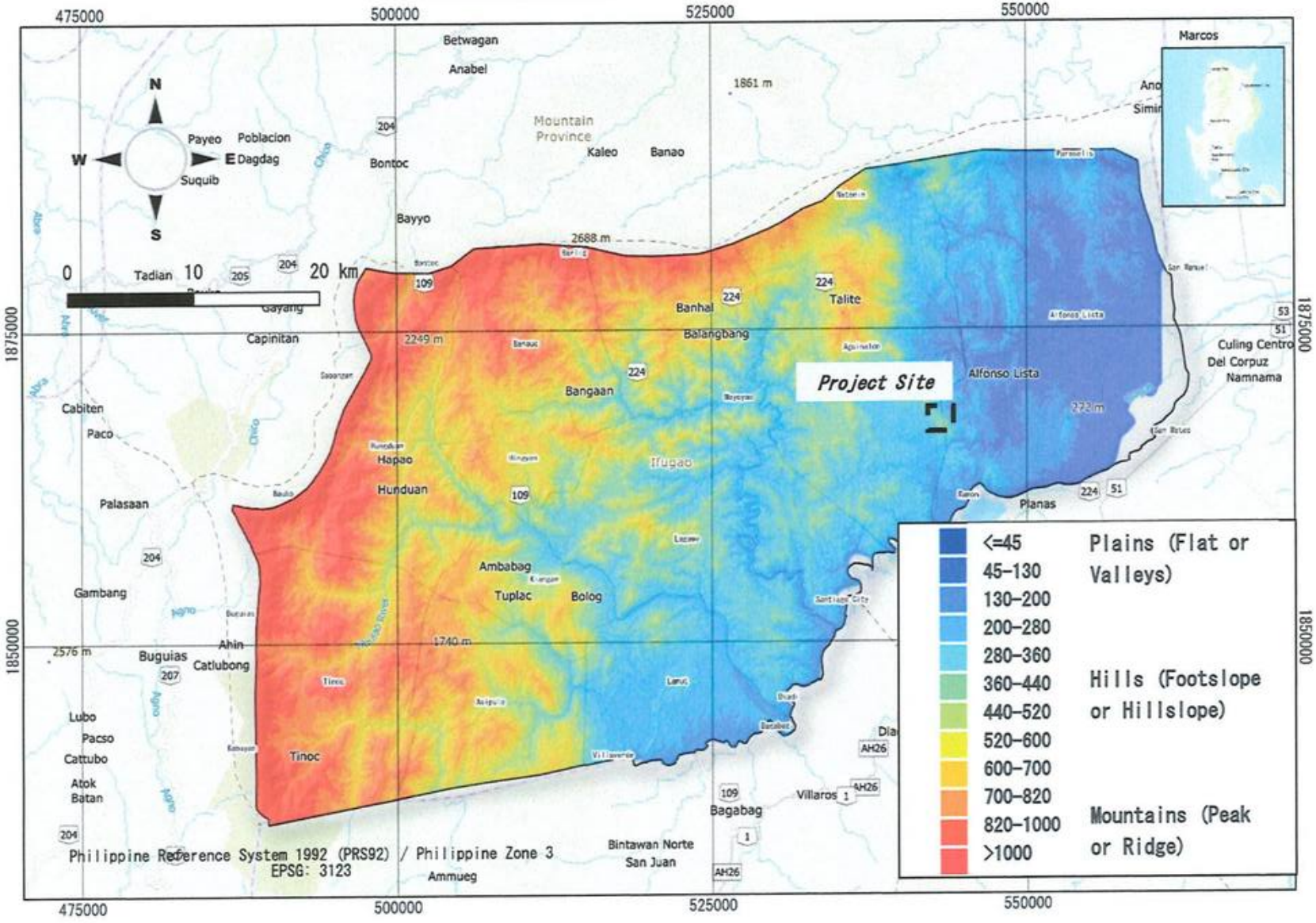
Earthquake-Induced Landslide Map of Ifugao



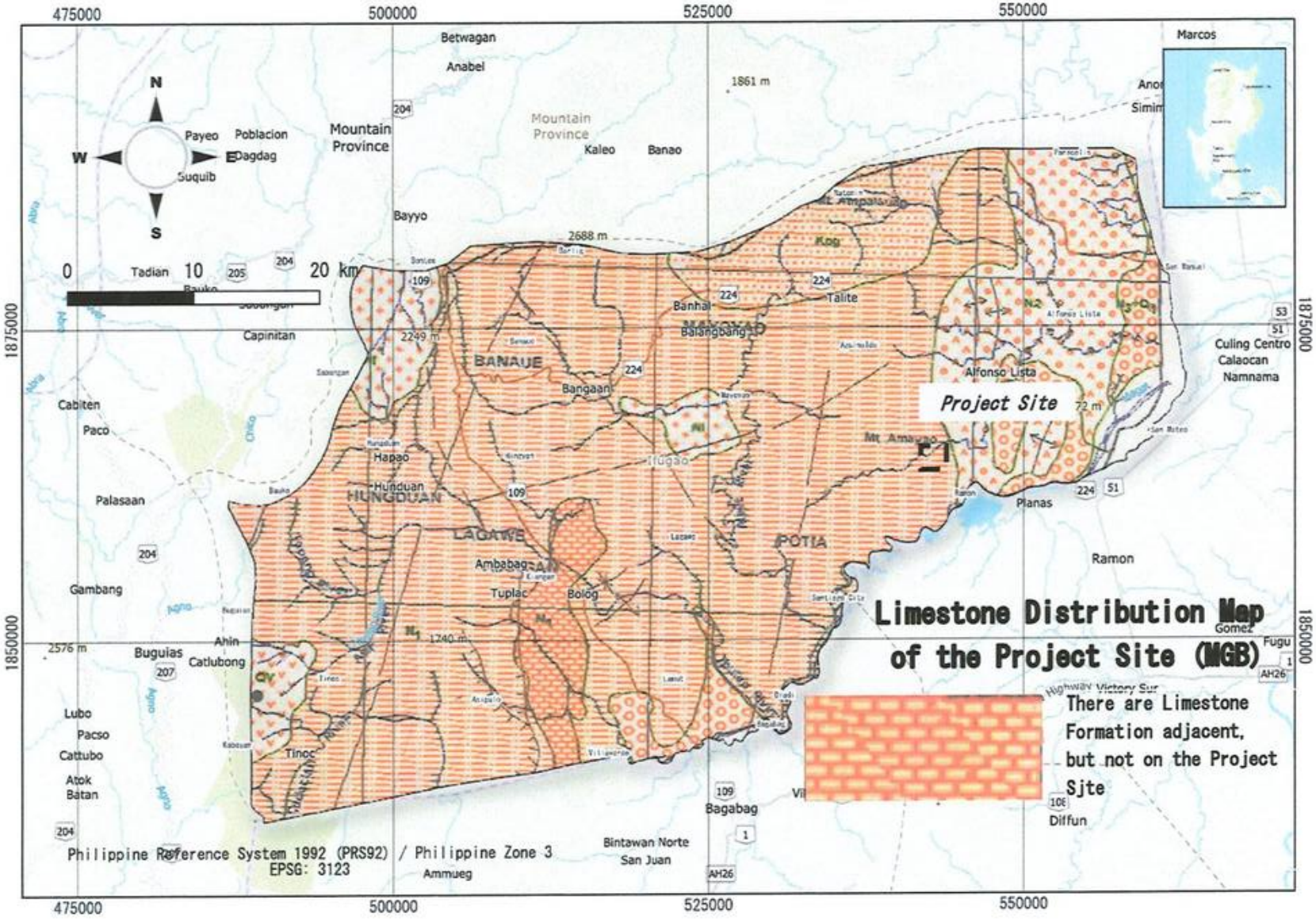
Geology Map of Ifugao



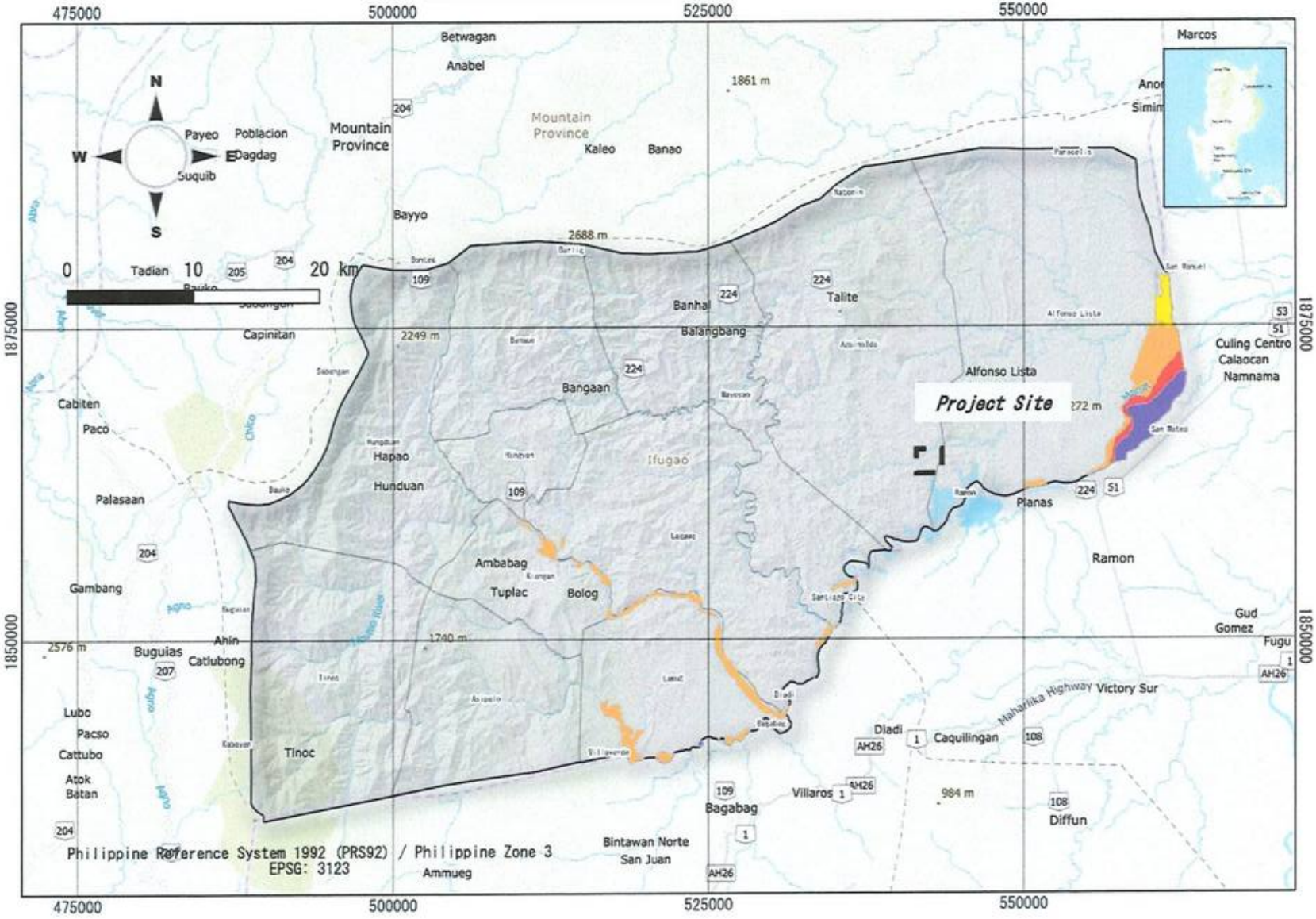
Geomorphology Map of Ifugao



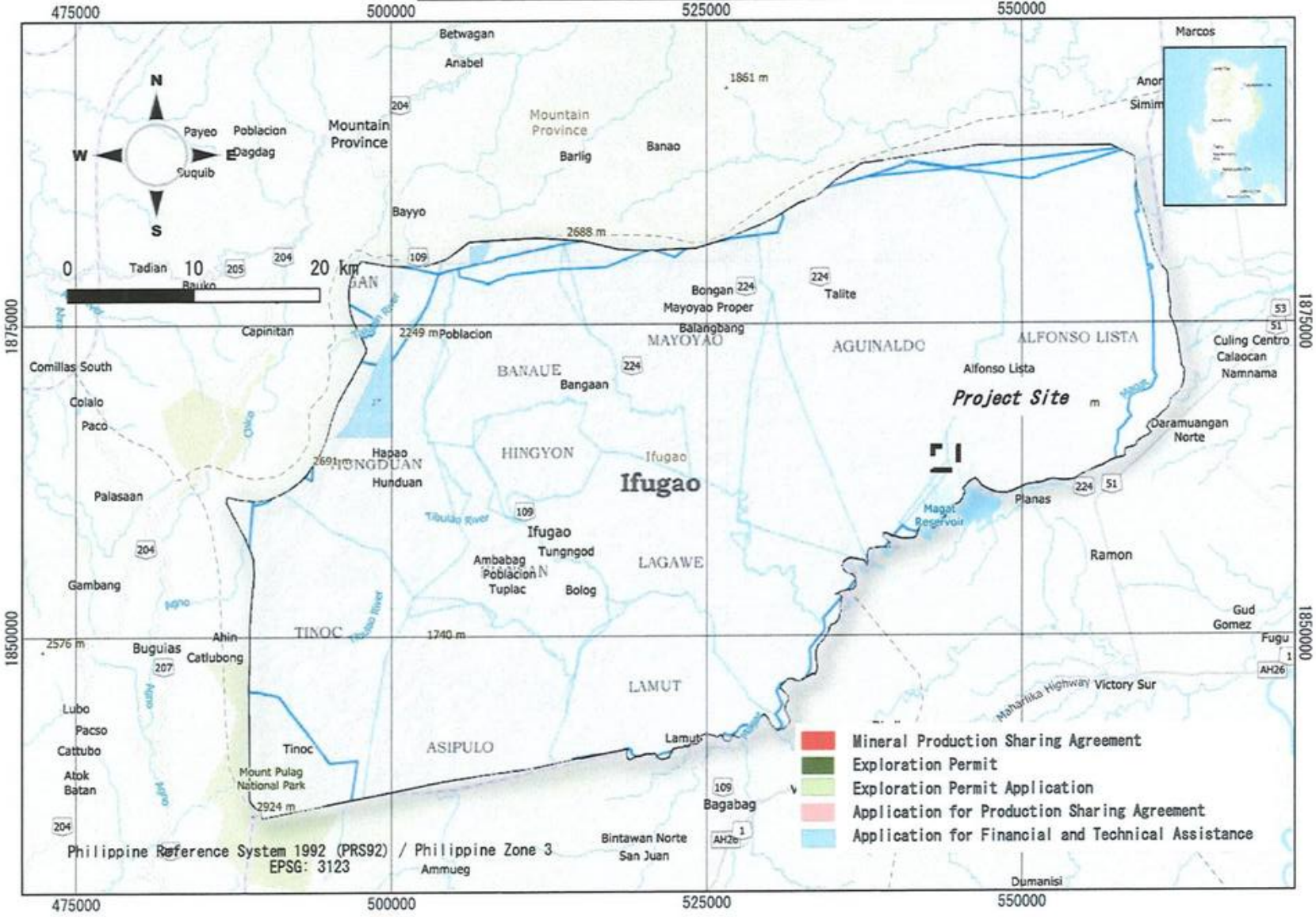
Karts (Limestone) Map of Ifugao

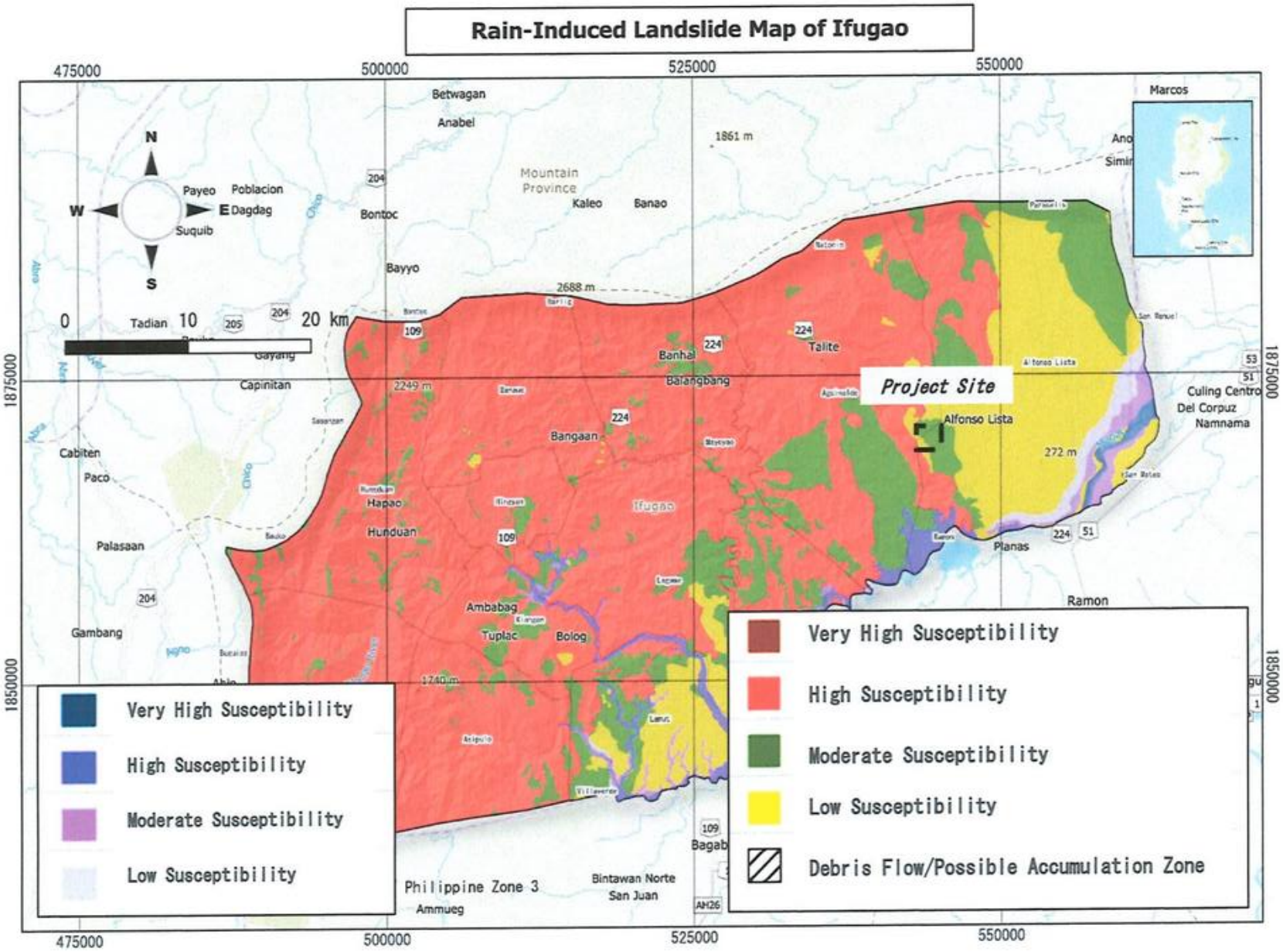


Liquefaction Map of Ifugao

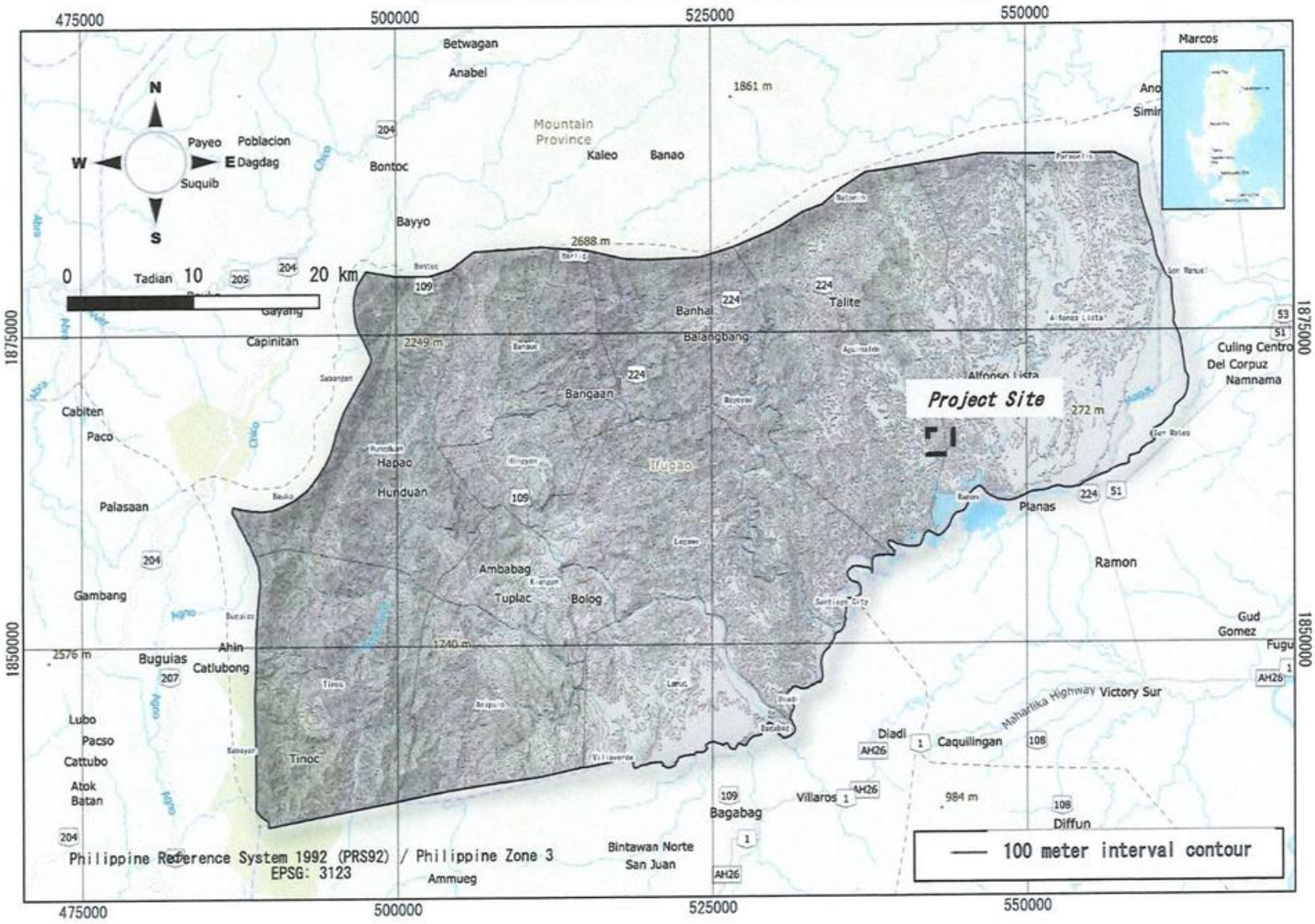


Mining Map of Ifugao

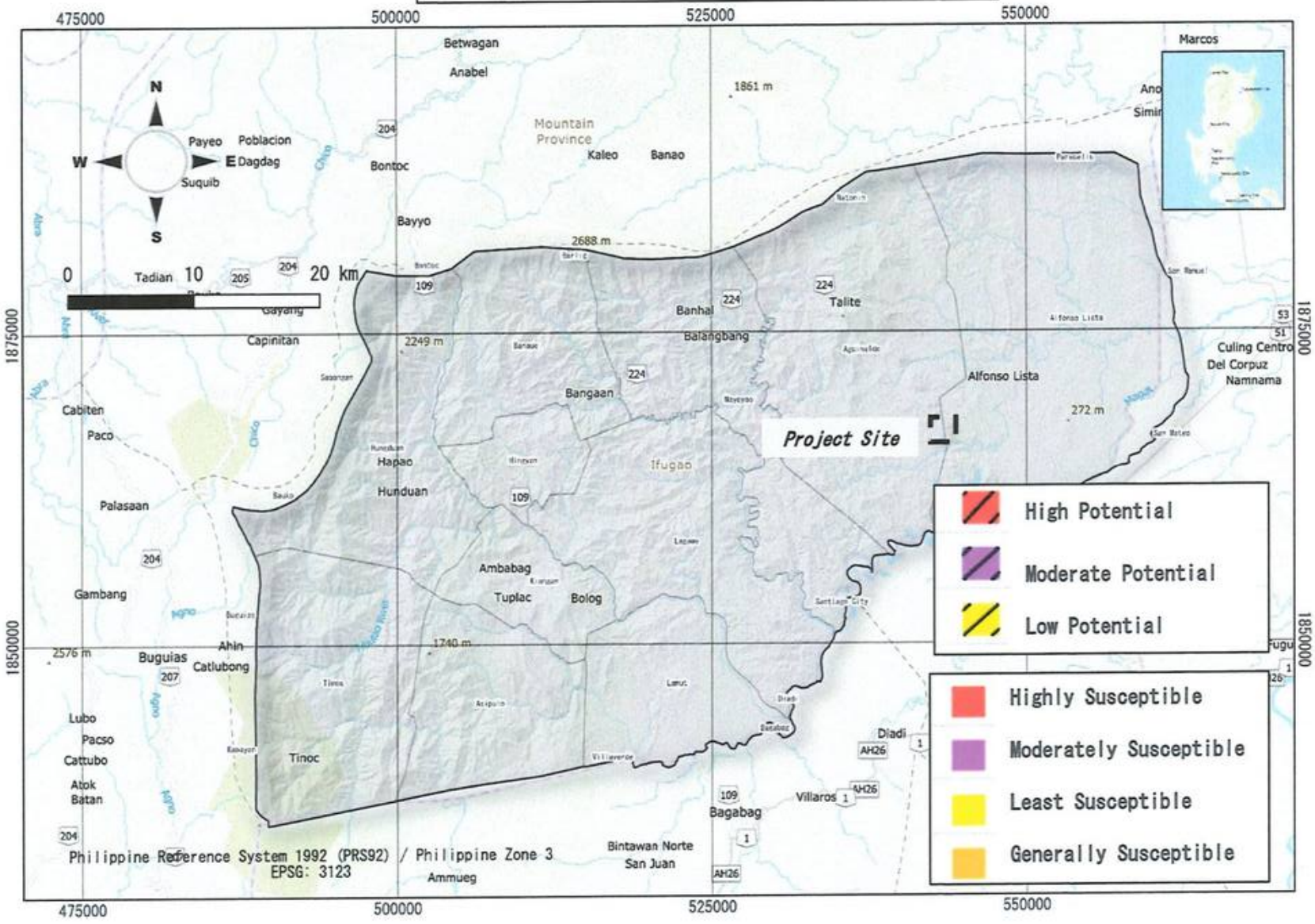



Rain-Induced Landslide Map of Ifugao
475000
500000
525000
550000
1875000
1850000
N
S
W
E
Payeo
Poblacion
Dagdag
Suquib
0
10
20 km
Betwagan
Anabel
Mountain Province
Kaleo
Banao
Bontoc
Bayyo
1861 m
2688 m
2249 m
1740 m
272 m
Tadian
Gayang
Capinitan
Cabiten
Paco
Palasaan
Gambang
Hapao
Hunduan
Ambabag
Tuplac
Bolog
Bangaan
Banhal
Balangbang
Talite
Alfonso Lista
Project Site
Planas
Ramon
Culling Centro
Del Corpuz
Namnama
Marcos
Ifugao
Philippine Zone 3
Ammueg
Bintawan Norte
San Juan
Bagab
Very High Susceptibility
High Susceptibility
Moderate Susceptibility
Low Susceptibility
Very High Susceptibility
High Susceptibility
Moderate Susceptibility
Low Susceptibility
Debris Flow/Possible Accumulation Zone

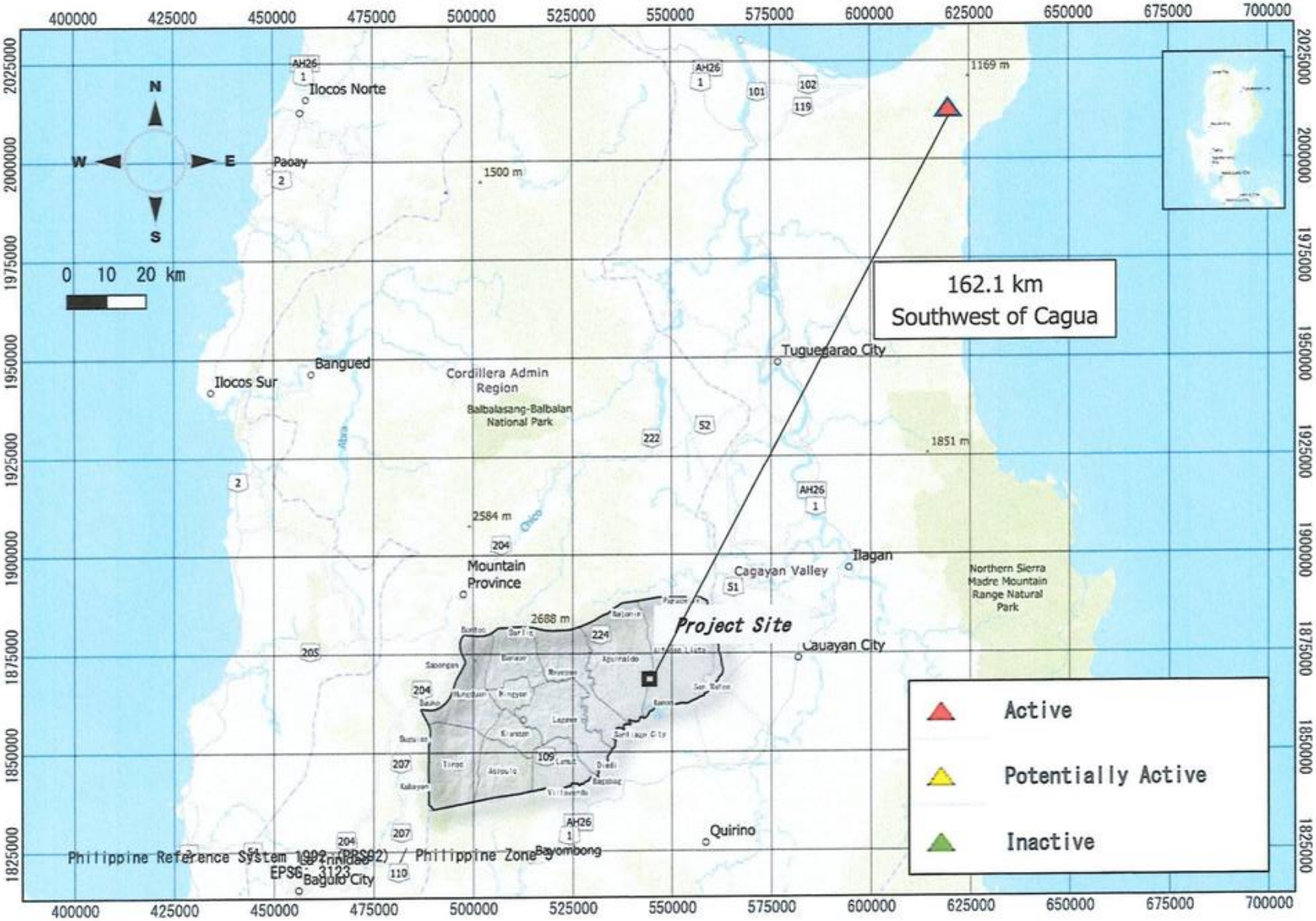
Topography Map of Ifugao



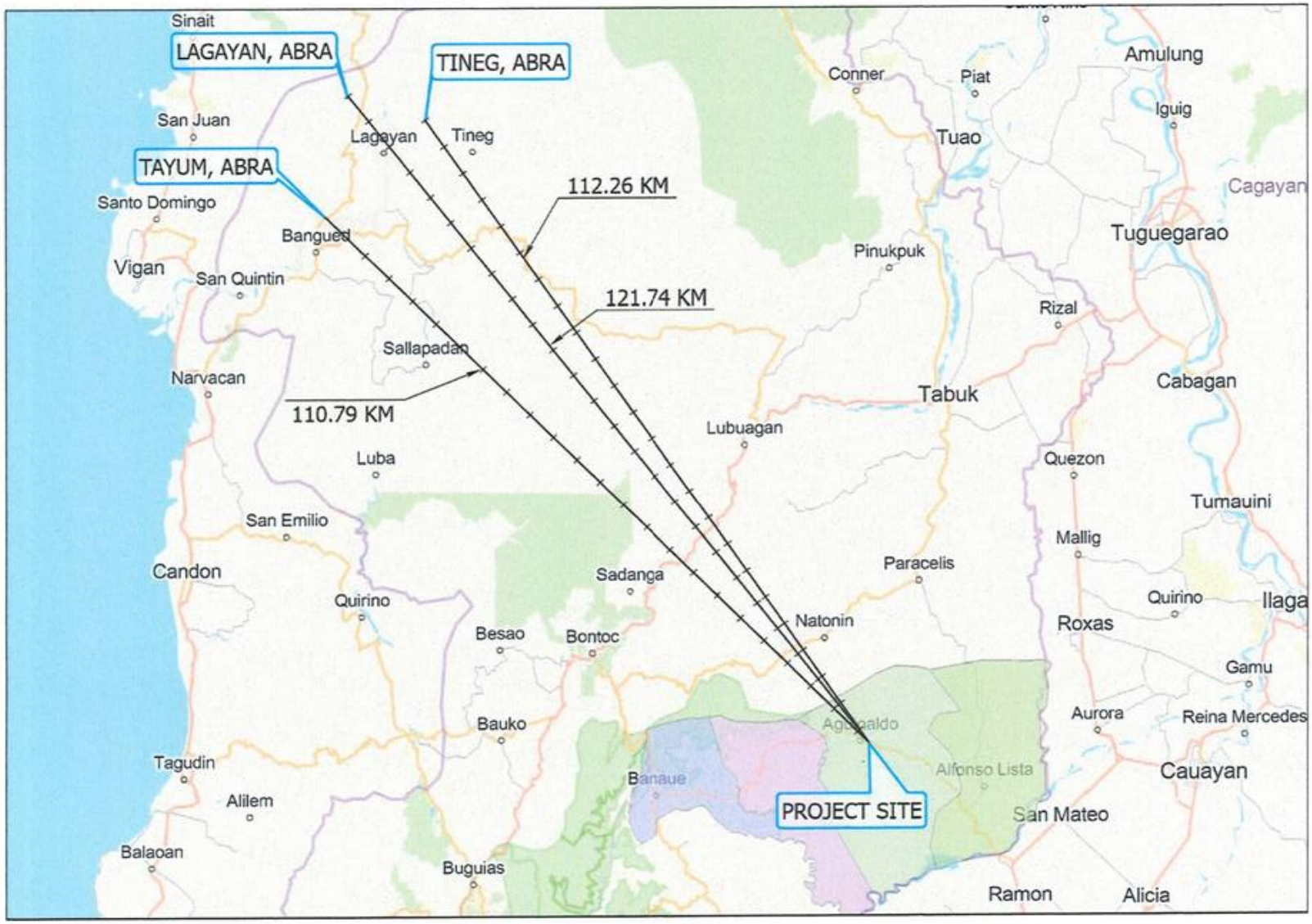
Tsunami-Surge Map of Ifugao



Volcano Map of Ifugao



Historical Earthquake Mapping



Project Site Photographs

